UNEP

OUR PLANET

The magazine of the United Nations Environment Programme - December 2007



SYMPHONY OF THE SEAS

The Marine Environment

OUR PLANET

Our Planet, the magazine of the
United Nations Environment Programme (UNEP)
PO Box 30552
Nairobi, Kenya
Tel: (254 20)7621 234
Fax: (254 20)7623 927
e-mail: uneppub@unep.org

To view current and past issues of this publication online, please visit www.unep.org/ourplanet

ISSN 101 - 7394

Director of Publication: Eric Falt
Editor: Geoffrey Lean
Coordinators: Naomi Poulton, David Simpson
Assistant Coordinator: Anne-France White
Special Contributor: Nick Nuttall
Distribution Manager: Manyahleshal Kebede
Design: Amina Darani
Producedy by: UNEP Division of Communications and Public Information
Printed by: Naturaprint
Distributed by: SMI Books

The contents of this magazine do not necessarily reflect the views or policies of UNEP or the editors, nor are they an official record. The designations employed and the presentation do not imply the expressions of any opinion whatsoever on the part of UNEP concerning the legal status of any country, territory or city or its authority or concerning the delimitation of its frontiers or boundaries.

* All dollar ($) amounts refer to US dollars.

also

page 3 reflections
page 7 verbatim and numbers
page 15 books
page 24 people
page 25 awards and events
page 29 www
page 30 products

Vice-Admiral **Conrad C. Lautenbacher,** Jr., Under Secretary of Commerce for Oceans and Atmosphere, USA...



...decribes how UNEP's Global Programme of Action for the Protection of the Marine Environment from Land-Based Activities assists sustainable management of oceans and coasts.

flexible instrument - page 4

Efthimios E. Mitropoulos, Secretary-General of the International Maritime Organization...



...describes what is being done to protect the marine environment from shipping.

ship shape - page 8

Prof Dr Ulf Riebesell of the Leibniz Institute of Marine Sciences in Kiel, Germany...



...describes how acidification of the oceans has become a new reason for rapid and decisive action to reduce carbon dioxide emissions.

acid oceans - page 10

Lee A. Kimball, a member of the Group of Experts of the Assessment of Assessments of the state of the marine environment...



...describes international efforts to build knowledge, through ocean assessment, as the basis of sound decision making.

building knowledge - page 12

His All Holiness Ecumenical Patriarch Bartholomew, the Archbishop of Constantinople...



...describes how disorder in the world's oceans and other waters is profoundly troubling for life on Earth.

in the same boat - page 16

Ibrahim Thiaw, Director of UNEP's Division of Environmental Policy Implementation...

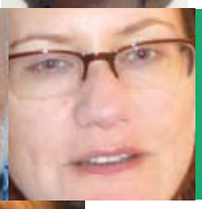


...describes how UNEP's Oceans and Coasts Programme is being redesigned in the face of new challenges.

rethink, realign, redirect - page 18

Gerald Marten, an ecologist at the East-West Center in Honolulu and author of *Human Ecology: Basic Concepts for Sustainable Development* and **Amanda Suutari,** an environmental journalist,...

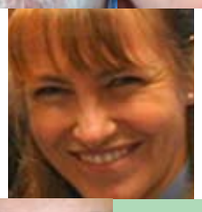




....describe how restoring mangroves can turn a vicious environment and development cycle into a virtuous one.

tipping points - page 20

Sandra Bessudo, Director of the Malpelo/ MarViva Foundation Colombia...



...describes a pioneering bid to conserve a unique patch of the world's seas.

living laboratory - page 22

Philippe Gautier, Registrar, International Tribunal for the Law of the Sea; Professor, Université Catholique de Louvain (Louvain-la-Neuve) ...



...describes the importance and role of the Law of the Sea in protecting and preserving the marine environment.

matters of judgement - page 26

Cameron Diaz, the actress who has appeared in 35 successful Hollywood films and was nominated four times for the Golden Globes...



... describes her green upbringing and explains how she encourages others to be environmentally-friendly.

everyone's cause - page 31

tipping points

by Gerald Marten and Amanda Suutari



Mangroves — the 'rainforests by the sea' — cover one-quarter of the Earth's tropical and subtropical coastline. Occupying two worlds, as the interface between land and sea, they are vital for healthy coastal ecosystems, which in turn support healthy fisheries — and, indeed, themselves act as nurseries for fish. They increase the resilience of coasts, protecting them from erosion, tropical storms and tidal waves. They trap sediment running off the land, safeguarding seagrass beds and coral reefs from siltation. And they function as a natural 'supermarket', providing necessary materials to those who depend on them — such as fruits, honey, other foods, fuelwood, medicinal plants and construction material. But they are also among the world's most threatened habitats and their rate of disappearance is accelerating with the conversion of coastal lands for development, charcoal production, tourism, and the controversial practice of shrimp aquaculture.

From 1975 to 1993 about half the mangroves along Thailand's 2,560 kilometre coast were lost. This had devastating effects, for example, in Trang province, some 800 kilometres south of Bangkok on the western shore of the Andaman Sea. Until the 1960s, its coastal villages largely subsisted on fishing while depending on the mangroves for medicinal plants and materials such as thatch for housing and fishing gear. Then the mechanization of fishing set into motion a range of effects, which seriously undermined the villages' natural and social capital. Large commercial trawlers violated the 3km coastal zone where the villagers fished. Their destructive gear and methods damaged coral, scraped the seabed, and cleared out young fish which had not yet reproduced, but villagers were afraid to confront them.

At much the same time, mangrove forests were opened up to concessionaires who began clearing them for charcoal production. Some of the poorest villagers saw no other option than to accept low-paid jobs cutting mangroves for concessionaires or on commercial trawlers, destroying their own resource base. Villagers also began clearing the mangroves themselves, with the attitude that 'if I don't cut them, someone else will'. Women began to look for unskilled, low-paid work in factories, leaving children behind with aging grandparents in the village, further undermining the social fabric.

As the fisheries declined — under the impact of both the trawlers and mangrove destruction — villagers had to go further out to sea, and resorted to more destructive methods to catch dwindling numbers of fish — such as using dynamite, cyanide and pushnets which scraped the ocean floor, damaging sea grass beds, coral reefs and other marine habitats. They also had to invest in more expensive equipment to 'keep up' with others in the race for dwindling fish. Some resorted to selling off land. The coastal communities were caught in a trap where day-to-day survival strategies eliminated or reduced their future options: the result was a self-reinforcing downward spiral into increasing poverty, and social and environmental degradation.

In 1985, a small organization called Yadfon ('raindrop' in Thai) began to work with the coastal villagers. Its founders, Pisit and Ploenjai Charnsnoh, talked for months with the people and the local imam, building confidence and identifying urgent needs. Together they dug a community well and set up a cooperative programme to help fishers buy fishing gear and engines for their boats, sell their daily catch at fair market prices, and reduce their dependence on middlemen. They established a revolving fund to give small interest-free loans to the poorest, most indebted villagers helping them to set up income-generating projects like small-scale aquaculture cultivating mussels, oysters, and grouper.

While all this was taking place, the villagers came up with the idea of reviving their badly degraded mangrove forests. A group of villages created a 235-acre community-managed



forest and sea-grass conservation zone, the first of its kind in Thailand. They initiated no-fishing areas, discouraged the use of cyanide and dynamite, and banned pushnets. Sea grass was replanted in a lagoon, and mangrove seedlings in degraded areas of the forest.

Now there are about ten such community-managed forests, ranging from 12 to 700 hectares, each managed by the group of villages that surround or depend on it. While each has its own rules of management, not one allows shrimp farms within its boundaries because there is general agreement that these endanger the mangroves. Over the years, the forests have begun regenerating, and fisheries have revived as a result. In 1992, Yadfon co-founded the Mangrove Action Project, an international network of some 800 conservation groups and academics from 60 countries working to promote mangrove conservation.

Creating the community forests and related projects began to transform attitudes among villagers who had forgotten traditional ways of working together, and helped them rediscover a sense of engagement, solidarity, and confidence. As their unity developed, leaders began to emerge, and newfound talents began to shine. Successes gave the people confidence that they had the power to help themselves instead of perceiving themselves as victims of an unfair system, waiting for government rescue. Building assets gave them a sense of ownership over their shared resource, and an incentive to band together to protect them from outside interests. Investing in their future motivated them to fight for it. Fishers began confronting trawlers who violated the 3km coastal zone and lobbied the government to enforce it. And when a local corporation spilled poisonous palm oil into a local waterway, villagers took the issue to provincial authorities, eventually forcing the company to pay compensation.

The creation of the mangrove forests was thus an Eco Tipping Point: a lever setting in motion a cascade of far-reaching effects that tipped the local community and environment from a vicious cycle to a virtuous one. The momentum switched from destruction to recovery and sustainability. The invasion of commercial interests into communal resources — the largely simultaneous mechanization of fishing and the arrival of charcoal concessionaires — was a negative tipping point that locked the people into a vicious cycle of declining resources and an accelerating race to get what little remained. Their prospects seemed so bleak that recovery would have seemed a fantasy.

But all this was reversed, and a virtuous cycle initiated, when villages began creating community mangrove forests and seagrass beds. The fishery began to restore itself, and the mangroves began to supply useful products again, reinforcing the community's commitment to protecting and managing them. Using simple wooden traps or nets, children can now earn 250-300 baht from catching crabs in the mangrove in an afternoon — the equivalent of what was once a whole a day's earnings cutting mangrove trees for concessionaires. Instead of being locked into depletion, villagers are locked into conservation, as the financial incentive to preserve mangroves now outweighs the incentive to destroy them. Similarly, in a study of 500 families from1991 to1994, the total fish catch rose by 40 per cent. And as the fishers spent 3-4 hours fewer in their boats and did not go out as far, their net income increased by 200 per cent. They could return with full boats without using dynamite or push nets. Fish stocks recovered faster, making their jobs even easier. And better incomes meant there was less need to migrate from villages.

Eco Tipping Points, such as these, offer a new paradigm for restoring communities, both natural and human. Conventional approaches to ecological problems — from piecemeal micromanagement, to techno-fixes, to top-down regulation — often fail. But with the right levers, the same forces that endanger environments and communities can be harnessed to heal them. UNEP